Ruilin Liu

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Summary

Ph.D. student in Natural Language Processing and Information Sciences, focusing on Large Language Model Interpretability and Agents. Research interests include computational linguistics and cognitive science. Experienced in leading research teams, constructing large-scale datasets, and developing advanced LLM-based systems, with strong expertise in agentic systems and interpretability/attribution methods.

Education

Nanjing Agricultural University — Nanjing, China
Ph.D. in Information Resource Management, College of Information Management — *Sep 2024 – Present*

- **Research focus**: NLP, LLM Reinforcement Learning, Mechanistic interpretability, Digital Humanities
- **GPA**: 3.84/4

Shandong Agricultural University, College of Information Science and Engineering — Taian, China
Master of Agricultural Engineering and Information Technology — *Sep 2022 – Jun 2024*

- **Key Courses**: Python, Data Mining, Machine Learning, Deep Learning, Data Structures, Probability & Statistics
- **GPA**: 3.77/4

Shandong Agricultural University — Taian, China
Bachelor of Science in Applied Chemistry — *Sep 2017 – Jun 2021*

Publications

- B. Li, B. Chang, **Liu, R.**, X. Zhao, S. Shen, L. Liu, Y. Zhu, Z. Xu, W. Qu, and D. Wang. Overview of EvaHan2025: The First International Evaluation on Ancient Chinese Named Entity Recognition. In *Proceedings of the Second Workshop on Ancient Language Processing*, pages 156–164, Albuquerque, New Mexico, 2025. Association for Computational Linguistics
- **Liu, R.**, D. Zhu, Y. Wei, and J. Song. From separation to entanglement: Dissecting feed-forward network functional organization in vision–language models for ancient chinese documents. *Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing*, 2026. EMNLP Findings
- **Liu, R.**, X. Guo, H. Zhu, and L. Wang. A text-speech multimodal Chinese named entity recognition model for crop diseases and pests. *Scientific Reports*, 15(1):5429, 2025
- X. Yao, X. Hao, **Liu, R.**, L. Li, and X. Guo. AgCNER: The first large-scale Chinese named entity recognition dataset for agricultural diseases and pests. *Scientific Data*, 11(1):769, 2024
- **Liu, R.**, Z. Zhao, Y. Ma, and D. Wang. Guji-tai: Task-aware interpretability for large language models in ancient chinese text processing. *Under Review*, 2026

Research Experience

SiNong Large Language Model Project — Nanjing Agricultural University
Researcher — *Nov 2025 – Present*

- Participated in building Sinong agricultural domain foundation model, collecting and organizing multi-source heterogeneous data across Animal Science, Smart Agriculture, Plant Protection, Crop Breeding and other sub-disciplines (> 4B tokens) to support vertical LLM training
- Designed high-quality agricultural instruction data pipeline, integrating synthetic data generation, instruction fine-tuning and reinforcement learning, and introducing Chain-of-Thought (CoT) and in-context reference QA formats to enhance agricultural reasoning and knowledge utilization
- Developed multi-agent Retrieval-Augmented Generation (RAG) framework optimized for domain literature retrieval, improving knowledge base construction, data source integration, and retrieval efficiency for professional agricultural documents
- Designed multi-agent architecture integrating multiple agricultural expert models, with intelligent routing and collaborative reasoning to dynamically coordinate domain-specific capabilities across Animal Science, Smart Agriculture, Plant Protection, and Crop Breeding

AI Companion Robot for Autism Spectrum Disorder (ASD) Nanjing Agricultural University
Team Leader *Aug 2025 – Mar 2026*

- Led multidisciplinary team ($n = 8$ members) developing human-machine emotional interaction system for individuals with Autism Spectrum Disorder, coordinating engineers, linguists, and clinical psychologists to ensure clinical reliability
- Constructed comprehensive domain-specific corpus ($> 50,000$ documents) from academic publications and specialized textbooks, structuring data to capture ASD behavioral patterns, communication strategies, and therapeutic approaches
- Developed Autism Companion LLM through pretraining → fine-tuning → RLHF pipeline using PyTorch and Hugging Face, achieving $> 85\%$ appropriateness rating in affective response generation and emotional understanding tasks
- Deployed model as core intelligent engine for interactive companion robot, processing all verbal communication and emotional interactions with individuals with ASD, enabling real-time dialogue with < 500ms response latency

Major Program of the National Social Science Fund of China Nanjing Agricultural University
Researcher *Sep 2024 – Present*

- Optimized Xunzi Large Language Model for classical Chinese comprehension through architectural analysis, designing reinforcement learning framework targeting polysemy, complex syntax structures, and deep cultural context unique to ancient texts
- Designed various tasks specific to the field of Classical Chinese, such as text translation, text matching, word segmentation, named entity recognition, and punctuation annotation.
- Designed comprehensive evaluation system combining manual expert assessment ($n = 3$ classical Chinese literature scholars) and automated metrics (BLEU, ROUGE, domain-specific accuracy), achieving a great improvement in ancient text comprehension
- Collaborated iteratively with domain experts to refine evaluation criteria based on academic feedback, enhancing model's cultural fidelity for research and educational applications while reducing anachronistic interpretations by $> 40\%$

NAACL ALP Workshop – Evahan 2025 Ancient Text NER Competition NAACL Conference
Organizer *Apr 2025 – May 2025*

- Organized and executed Evahan 2025 Competition during NAACL conference in Albuquerque, attracting many participating teams to benchmark ancient Chinese Named Entity Recognition methods
- Designed competition framework targeting unique linguistic features of ancient texts (polysemous characters, special word combinations, contextual variations), creating diverse test scenarios examining technical depth and innovation
- Coordinated high-quality dataset preparation with preprocessing for usability and challenge balance, developing multi-scenario test sets ($n = 8,000$ annotated entities) covering historical figures, place names, book titles, allusions, and polysemantic entities
- Established evaluation standards ensuring reproducibility and managed end-to-end workflow including participant assessment pipeline and workshop publication, creating new benchmark for ancient text NER research

Shandong Provincial Natural Science Foundation Youth Program Shandong Agricultural University
Researcher *Oct 2022 – May 2023*

- Constructed large-scale agricultural pest and disease text dataset with $> 70,000$ annotated samples achieving industry-leading annotation depth (avg. 3.4 entities per sample), establishing comprehensive resource for agricultural NLP research
- Independently developed multimodal text-speech dataset ($n = 20,000$ annotated samples, $68,513$ labeled entities) for agricultural pest and disease diagnosis, creating first benchmark dataset for Chinese agricultural multimodal Named Entity Recognition
- Proposed novel text-speech multimodal NER approach integrating acoustic and textual features, achieving 91.2% F1-score for pest and disease entity extraction and outperforming text-only baseline by 12.8%
- Published dataset and methodology in Scientific Reports and Scientific Data (Nature portfolio), providing foundational resources that enable subsequent agricultural NLP research and practical diagnostic applications

Technical Skills

Web Frameworks & AI Coding Tools	Django, Flask, Angular, Vue.js, Claude Code
NLP & Deep Learning	LLM Pre-training, Fine-tuning, RLHF, RAG, Multimodal Learning
Data & Annotation	Corpus Construction, Annotation Pipeline Design, Data Preprocessing
Database & Data Storage	SQL (MySQL, PostgreSQL), MongoDB
Personal Website	https://liu-ruilin.github.io/
Project Outcomes	https://xunziallm.njau.edu.cn/
	https://nongzheng.njau.edu.cn/
Computing Infrastructure	Freely available for 8*NVIDIA A800 (80GB)